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RESEARCH ARTICLE

GKit-SSRDecoder: An Open-Source C/C++-Based PPP-B2b and HAS Decoding and Formatted Product Generation Software

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ABSTRACT With the rapid development of Real-Time Precise Point Positioning (RT-PPP), the BeiDou-3 PPP-B2b and Galileo High Accuracy Service (HAS) have become pivotal for providing free, global, satellite-based precise correction services. However, most existing open-source decoding tools are developed based on the Python language, which limits their integration into embedded systems and real-time C/C++ positioning frameworks. In addition, these raw data streams lack standardized input formats and product generation, and do not take into account user-side satellite antenna calibration, which hinders offline analysis and cross-platform verification. To bridge these gaps, this study develops and presents the GKit-SSRDecoder, an open-source, high-performance C/C++-based software designed for the end-to-end processing of PPP-B2b and HAS messages. The software features a modular architecture that supports multi-receiver raw data stream parsing, State Space Representation (SSR) message decoding, and the automated generation of RINEX-compliant precise orbit, clock, and code bias products. Comprehensive evaluations were conducted using data from diverse GNSS receivers and compared against post-processed products. The results demonstrated that the Signal-In-Space Range Errors (SISRE) for the PPP-B2b service is 0.093 m and 0.086 m for GPS and BDS-3, respectively, while the HAS service achieves 0.057 m for GPS and 0.048 m for Galileo. Positioning tests showed that the generated products enable RT-PPP to reach centimeter-to-decimeter-level accuracy in static and kinematic modes. Specifically, in real-time kinematic scenarios, the positioning performance of the PPP-B2b and HAS services achieves RMS values of 0.13/0.11/0.12 m and 0.18/0.12/0.23 m in the East/North/Up directions, respectively. This software serves as a useful and portable tool for the GNSS community, facilitating the industrial application and scientific research of satellite-based augmentation services.

INDEX TERMS Real-time precise point positioning, PPP-B2b, high accuracy Service, open-source software.

I. INTRODUCTION

Precise Point Positioning (PPP) technology has been widely applied in Positioning, Navigation, and Timing (PNT) applications owing to its characteristics of all-weather adaptability and global high-accuracy services [1]. Initially, PPP was generally performed in post-processing mode because of the latency of precise orbit and clock products [2], [3]. To satisfy the demand for real-time PPP (RT-PPP) applications, many analysis centers or industrial service providers provide real-time orbit and clock corrections according to the Radio Technical Commission for Maritime Services (RTCM) protocols [4], [5]. These corrections are broadcasted to global users via the Internet based on Networked Transport of RTCM via Internet Protocol (NTRIP) with very low latency [6], [7], [8], [9].

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However, these internet-dependent ground-based services are inaccessible in remote areas like deserts or oceans. To address this, satellite-based services such as the BDS-3 PPP-B2b and the Galileo High Accuracy Service (HAS) service have emerged, broadcasting precise corrections directly via satellite signals [10], [11], [12]. While these services successfully enable internet-free RT-PPP, they broadcast corrections in proprietary binary formats due to bandwidth constraints, creating significant interoperability barriers for users. To overcome these decoding barriers and facilitate scientific applications, the specific aim of this research is to provide an efficient, open-source C/C++ software, namely GKit-SSRDecoder, for decoding and formatting PPP-B2b and HAS messages from diverse receiver manufacturers, ensuring scientifically assessed service accuracy and seamless integration into real-time precise positioning applications.

The rest of this paper is organized as follows: Section II reviews related works regarding satellite-based services and existing decoding tools. Section III introduces the framework and workflow of the GKit-SSRDecoder software. Section IV evaluates the quality of the decoded B2b/HAS products and assesses the real-time PPP performance. Finally, conclusions are drawn in Section V.

II. RELATED WORK

A. SATELLITE-BASED SERVICES ADVANCEMENT

With the operationalization of satellite-based correction services, extensive research has evaluated their performance. the PPP-B2b service directly broadcasts precise orbit, clock, and Differential Code Biases (DCB) corrections via B2b signal of three Geostationary Earth Orbit (GEO) satellites, enabling real-time PPP for users within China and its surrounding regions [10]. The Galileo HAS service of the European Union provides orbit, clock, code, and phase bias in Observable-specific Signal Biases (OSB) format worldwide for free through its E6 signal [11]. Besides, the QZSS Centimeter Level Augmentation Service (CLAS) provides comprehensive State-Space Representation (SSR) corrections over Japan, enabling rapid carrier-phase ambiguity resolution within minutes [13]. Several studies have comprehensively analyzed and compared the feasibility of those services from various perspectives [14], [15], [16], and successfully applied them to scientific research [17], [18]. Moreover, specific attention has been paid to the strategies for estimating PPP-B2b phase bias products, developing novel receiver clock models, and combining different services [19], [20], [21]. These evaluations demonstrate that those services can achieve decimeter- to centimeter-level positioning accuracy, satisfying the stringent requirements for real-time precise positioning.

B. CURRENT LIMITION AND MOTIVATION

Despite the proven accuracy of PPP-B2b and HAS services, the broadcasting of corrections in proprietary binary formats introduces significant interoperability barriers due to heterogeneous frame headers across manufacturers. While several open-source decoding tools have emerged as summarized in Table 1, they face critical technical bottlenecks. For instance, HASlib is restricted to specific input formats, while GHASP yields outputs that deviate from standard PPP requirements [22], [23]. More importantly, the predominant reliance on Python in frameworks like NavDecode and CSSRlib [24], [25] creates a significant gap. While Python is highly efficient for post-processing analysis, it introduces substantial latency and integration challenges when embedded into low-power, mainstream C/C++ positioning frameworks. Although HASPPP offers a C/C++ alternative for the HAS service [26], it currently lacks comprehensive handling of satellite antenna phase center (APC) corrections as well as the absence of ephemeris output.

TABLE 1. The status of current open-source software packages.

Software	Services	input	output	language
HASlib	HAS	raw binary data	RTCM 3/IGS SSR	Python
GHASP	HAS	raw binary data	CSV	Python
HASPPP	HAS	raw binary data	products cached by BNC	C/C++
NavDecode	HAS/PPP-B2b	converted text data	ASCII	Python
CSSRlib	HAS/PPP-B2b	converted text data	NO	Python

Therefore, a significant research gap remains: existing tools are often implemented in Python, which, while suitable for post-processing, poses challenges for low-latency, real-time integration into industrial C/C++ positioning engines. Furthermore, the lack of a unified interface for multi-vendor hardware protocols necessitates a redundant development process. The GNSS community urgently requires a unified, full-featured, C/C++ based end-to-end stream decoding software that natively supports both PPP-B2b and HAS multi-vendor protocols, which is the primary motivation for developing GKit-SSRDecoder.

III. METHODOLOGY

The GKit-SSRDecoder supports the decoding of raw binary B2b/HAS data from various manufacturers and outputting precise ephemeris and DCB/OSB products in Receiver Independent Exchange Format (RINEX), and the reference center of the orbit products could be customized by users. The methodology of GKit-SSRDecoder is built upon a modular streaming architecture. It involves four critical layers: (1) data input and reading; (2) SSR message decoding; (3) satellite precise ephemeris calculating; (4) result and product outputting. The detailed flowchart of the GKit-SSRDecoder is presented in Fig. 1.

A. DATA INPUT AND READING

The datasets required by GKit-SSRDecoder are divided into mandatory and optional components, the mandatory data include the raw binary files of B2b/HAS, satellite antenna correction (.atx) files, and broadcast ephemeris files in

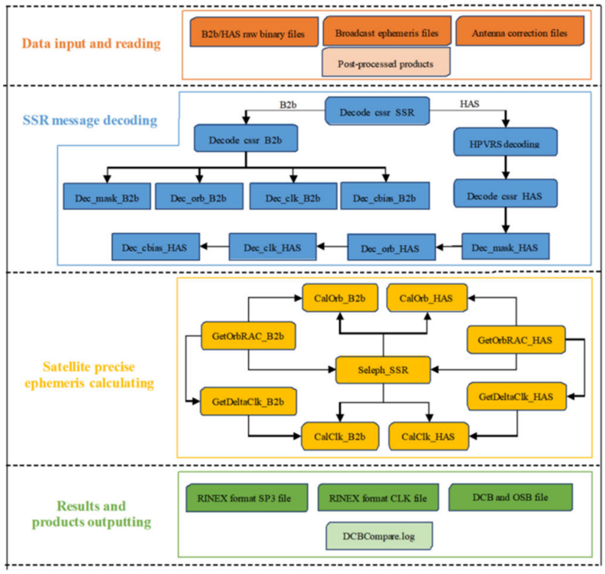

FIGURE 1. Architecture of the GKit-SSRDecoder software.

RINEX 4.0 format. It is optional to input DCB files from the Chinese Academy of Sciences (CAS) or Center for Orbit Determination in Europe (CODE) for assessing the accuracy of code biases broadcast by PPP-B2b and HAS services. It should be noted that the antenna file must be consistent in date with the B2b/HAS data, and the HAS service always uses version 14. Otherwise, the accuracy of the recovered satellite orbit products at the phase center will be affected by the inconsistency between them [27]. Moreover, broadcast ephemeris files from three consecutive days are needed when processing data over an entire day due to the IOD version delay in the B2b/HAS service. Note that all files need to be placed in the same file directory.

B. SSR MESSAGE DECODING

The GKit-SSRDecoder supports raw binary data of B2b/HAS corrections collected by different types of receivers, which is conducive to the development of real-time software. By recognizing the custom data frame headers of various manufacturers, it extracts the raw B2b/HAS binary data for decoding. The extracted raw B2b data can be decoded based on their message types. In contrast, the extracted raw HAS C/NAV data needs to be further encoded with a High-Parity Vertical Reed-Solomon (HPVRS) scheme before decoding the corrections according to their message types [28], [29]. Currently, the software supports raw HAS data collected from Septentrio, NovAtel, Chinese SINO and Unicore receivers, as well as raw B2b data collected from Septentrio and Unicore receivers. Users can easily extend the capability to process data from other manufacturers.

C. SATELLITE PRECISE EPHEMERIS CALCULATING

The decoded B2b/HAS information in the previous step is the corrections with respect to the broadcast ephemeris rather than the raw satellite orbit and clock information, i.e., the corrections cannot be applied standalone without broadcast ephemeris. Initially, the approximate satellite position vector (r), satellite velocity vector (v), and clock offset (t) are calculated using the broadcast ephemeris, with GPS, Galileo, and BDS-3 employing LNAV, INAV, and CNAV1 navigation messages, respectively. It is worth mentioning that the calculation of satellite precise orbit and clock information for B2b and HAS differs. Specifically, the formulas are as follows:

$$
\begin{cases}
\widehat{r}_{B2b} = r - [\ \frac{r}{|r|}\ \ \frac{r\times v}{|r\times v|} \times \frac{r}{|r|}\ \ \frac{r\times v}{|r\times v|}\ ] \cdot \delta O_{B2b}, \widehat{t}_{B2b} \\
\qquad = t - \dfrac{\Delta t_{B2b}}{c} \\
\widehat{r}_{HAS} = r + [\ \frac{r\times v}{|r\times v|} \times \frac{r}{|r|}\ \ \frac{r}{|r|}\ \ \frac{r\times v}{|r\times v|}\ ] \cdot \delta O_{HAS}, \widehat{t}_{HAS} \\
\qquad = t + \dfrac{\Delta t_{HAS}}{c}
\end{cases}
\tag{1}
$$

where δO represents the decoded three-dimensional orbit corrections composed by the radial, tangential, and normal components, and Δt refers to clock offset correction of B2b/HAS; $c$represents the speed of light; $\widehat{r}$ and $\widehat{t}$ are the satellite precise orbit coordinates and clock offsets referred to satellite APC. Moreover, the reference APC varies across different systems. For GPS and Galileo systems, the satellite APC is associated with the dual-frequency ionosphere-free signal combinations of L1/L2 and E1/E5b, respectively. In contrast, the APC corresponding to BDS-3 satellites refers to the B3I signal. Therefore, to facilitate the applicability of the undifferenced and un-combined model, as well as incorporating other frequencies in real-time PPP, it is necessary to transform $\widehat{r}$ into satellite coordinates $r_s$referenced to the center of mass (CoM):

$$
\begin{aligned}
r_s^C &= \widehat{r^C} - pco_{\mathrm{B3I}} \\
r_s^G &= \widehat{r^G} - (\alpha_{\mathrm{L1/L2}} \cdot pco_{\mathrm{L1}} + \beta_{\mathrm{L1/L2}} \cdot pco_{\mathrm{L2}}) \\
r_s^E &= \widehat{r^E} - (\alpha_{\mathrm{E1/E5b}} \cdot pco_{\mathrm{E1}} + \beta_{\mathrm{E1/E5b}} \cdot pco_{\mathrm{E5b}}) \\
e_x &= \frac{e_y \times e_z}{\left|e_y \times e_z\right|}, e_y = \frac{\widehat{r} \times (\widehat{r} - r_{\mathrm{SUN}})}{|\widehat{r} \times (\widehat{r} - r_{\mathrm{SUN}})|}, e_z = -\frac{\widehat{r}}{|\widehat{r}|} \\
pco_j &= (\, e_x\ e_y\ e_z \,) \cdot (\, \Delta x\ \ \Delta y\ \ \Delta z \,)_j^{\mathrm{T}}
\end{aligned}
\tag{2}
$$

where r_{SUN} is the sun coordinates in Earth-center-Earth-fixed (ECEF) frame, α and β are dual-frequency ionosphere-free combination factor; $(\, \Delta x\ \ \Delta y\ \ \Delta z \,)_j$ represents the Phase Center Offset (PCO) corrections of frequency j. Similarly, the transformation of orbit product referenced to the combined double-frequency is the inverse process described above.

D. RESULTS AND PRODUCT OUTPUTTING

Subsequent to the aforementioned procedures, the software produces precise ephemeris products in RINEX format encompassing both SP3 and CLK files, at a time interval preset by the command parameters. Concurrently, the software outputs daily DCB and OSB files of all satellites, containing a single value for each. The format of DCB/OSB files adheres to the SINEX bias format, and the specific frequency channels included are listed in Table 2. The GPS C2W signal, which is not broadcasted in the HAS service, can be converted and output by the software using DCB products published by CODE.

The software is configured using command parameters in the format: "<YOUR DATAPATH><SSR Type><Time Interval><Time of Start><Time of End>". Here, <YOUR DATAPATH> is the directory for input data files. <SSR Type> can be set to "B2b" or "HAS" to identify raw SSR data files. <Time Interval> sets the output interval (in minutes) for SP3 and CLK files. <Time of Start> and <Time of End> define the data processing duration in the format of YYYY-MM-DD-HH-MM-SS. Moreover, the combination of frequencies appended to the command can specify the reference center of the orbit product, the configurable frequencies are shown in Table 2.

TABLE 2. The frequency channels of outputted B2b/HAS DCB/OSB products.

Service	frequency	channel
HAS	GPS: L1/L2	GPS: C1C/C2P/C2L
	Galileo:E1/E5a/E5b/E6	Galileo:C1C/C5X/C7X/C6C
B2b	BDS-3: B1I/B3I/B1C/B2a/B2b	BDS-3: C2I/C6I/C1X/C1P/C5X/C5P/C7Z/C7D

IV. ASSESSMENT OF THE DECODED B2b AND HAS PRODUCTS

To demonstrate the correctness of the GKit-SSRDecoder, the decoded B2b and HAS products were comprehensively assessed. The raw B2b and HAS binary data used in the experiment were sourced from Septentrio and China SINO receivers during day of year (DOY) 90-100 in 2025. The final GNSS precise orbit and clock products provided by Wuhan University (WUM) were employed as references to evaluate the accuracy of the PPP-B2b and HAS orbit and clock corrections. The performance of B2b/HAS real-time orbit and clock correction messages decoded by GKit-SSRDecoder was evaluated through the "SP3 Comparison" module in BNC software [30].

A. EVALUATION OF THE ACCURACY OF DECODED B2B PRODUCTS

Fig. 2 depicts the orbit (along-track, cross-track, and radial components) and clock offset differences of GPS and BDS-3 satellites between PPP-B2b and WUM. Most of the orbit and clock errors of GPS satellites are within 0.5 m, and the along-track component has the largest orbital errors while the radial component exhibits significantly smaller errors. Moreover, the GPS satellite clock errors are generally within 0.25 m. In terms of BDS-3 satellites, significant instabilities are observed in along-track and cross-track components, and the orbit errors of the three components are larger compared to those of GPS satellites. In contrast, BDS-3 satellite clock errors are smaller, indicating higher stability than the orbit components.

While most existing studies analyze 3D orbit errors and the signal-in-space ranging error (SISRE) solely from a statistical perspective, this study employs the methodology proposed

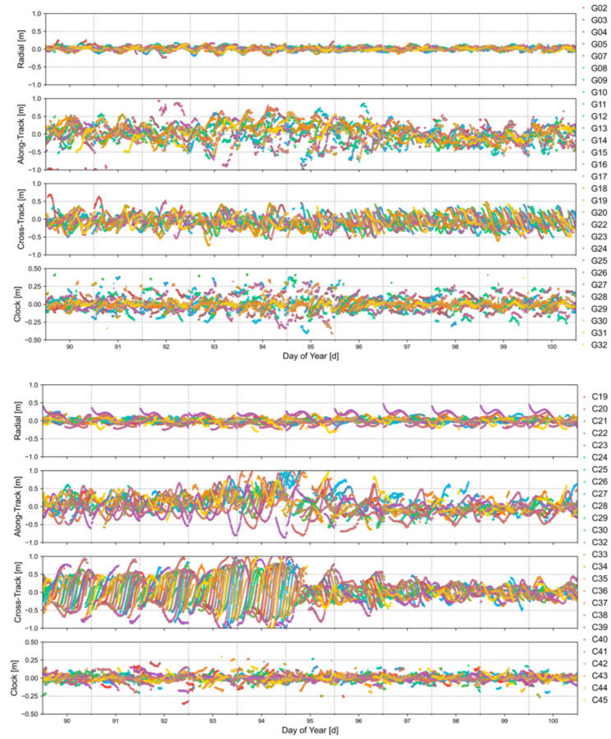

FIGURE 2. PPP-B2b orbit and clock errors from DOY 90-100 in 2025.

in [31] to calculate the root-mean-square (RMS) of weighted averaged orbit errors and SISRE considering the correlations between orbit and clock errors. Fig. 3 shows the RMS values of PPP-B2b orbit errors, clock errors, and SISRE for different satellites over a period of 11 days, and the corresponding averaged statistics for BDS-3 and GPS constellations are summarized in Table 3. The RMS of orbit and clock errors for BDS-3 satellites are 0.075 m and 0.051 m, with the IGSO satellites (C38, C39, C40) showing significantly larger orbital RMS values. In comparison, the GPS satellites exhibit a smaller orbit RMS of 0.054 m, but a larger clock RMS of 0.080 m compared with BDS-3. Overall, in terms of the SISRE, BDS-3 achieves a superior performance of 0.086 m, outperforming the values of 0.093 m for GPS.

TABLE 3. The average RMS of GPS/BDS-3 PPP-B2b orbit, clock, and SISRE errors (unit: m).

Satellite	Orbit	Clock	SISRE
GPS	0.054	0.080	0.093
BDS-3	0.075	0.051	0.086

The DCB corrections decoded by GKit-SSRDecoder software are compared with the products from CAS. Notably, the PPP-B2b service broadcasts DCB corrections for each frequency with respect to the B3I signal. Therefore, the OSB values from the CAS products must be converted into differential values with respect to the B3I reference frequency before comparison. Additionally, due to the differences in the reference frames, the differences for all satellites need to be adjusted by removing the mean offset. Fig. 4 summarizes the mean difference and standard deviation (STD) of PPP-B2b

FIGURE 3. RMS values of PPP-B2b orbit, clock, and SISRE errors.

code biases over a period of 11 days, with the corresponding statistical value calculated in Table 4. The accuracy of the BDS-3 B2b signal is the highest among all signals, with mean values of 0.241 ns and 0.260 ns for C7Z and C7D channels, respectively. Moreover, the accuracy of the BDS-3 B2a signal is 0.311 ns and 0.313 ns for C5X and C5P, respectively, which is almost comparable with the B2b signal. Besides, the B1C signal of BDS-3, with an accuracy of 0.424 ns and 0.419 ns for C1X and C1D, presents better performance compared to the B3I signal whose accuracy is 0.516 ns. In addition, the standard deviation of B1C code biases is close to that of B2b both falling within the range of 0.13 ns to 0.14 ns. Meanwhile, the B2a code biases show good stability over 11 days with STD of 0.090 ns and 0.074 ns.

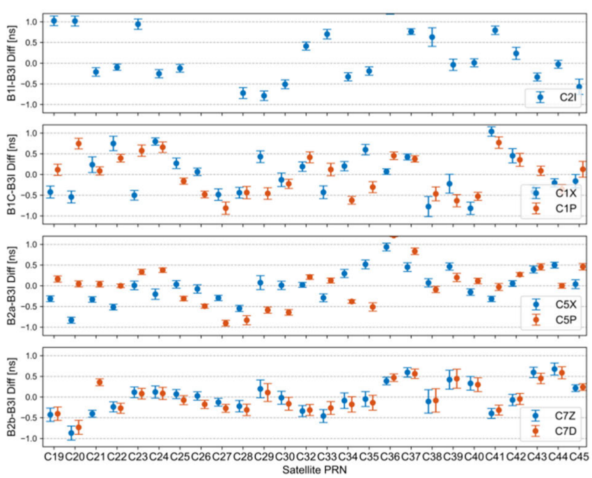

FIGURE 4. Mean difference and standard deviation of PPP-B2b code biases relative to CAS products.

To verify the superiority of the proposed method over existing approaches, the proposed GKit-SSRDecoder and published CSSRlib were employed to decode the same set of PPP-B2b data [24], [25]. The decoded BDS-3 orbit and clock products were compared against post-processed products, and the results are illustrated in Fig. 5 and 6. It is evident that the clock products decoded by both methods exhibit no significant difference. However, the three-dimensional orbit error decoded by this software is closer to zero, especially in the radial direction. The statistical results in Table 5 indicate that not considering the satellite-end APC correction will result in a deviation of more than 1 m in the radial direction.

TABLE 4. The average difference and standard deviation of PPP-B2b code biases (unit: ns).

Signal	C2I	C1X/P	C5X/P	C7Z/D
Diff	0.516	0.424/0.419	0.311/0.313	0.241/0.260
STD	0.102	0.140/0.133	0.090/0.074	0.137/0.137

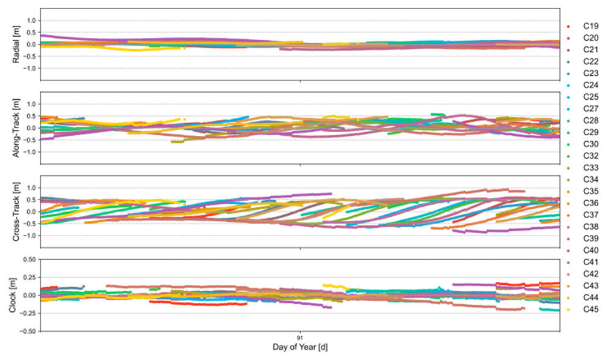

FIGURE 5. PPP-B2b orbit and clock errors on DOY 91 for GKit-SSRDecoder.

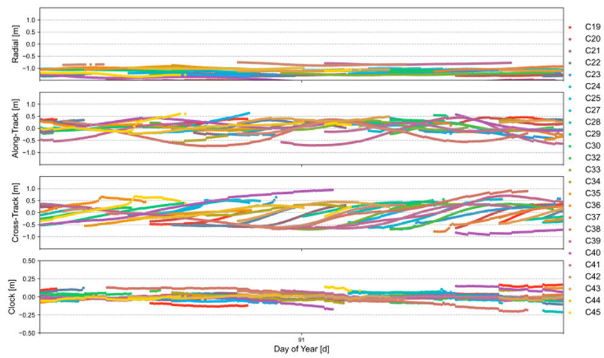

FIGURE 6. PPP-B2b orbit and clock errors on DOY 91 for CSSRlib.

TABLE 5. The average RMS of BDS-3 PPP-B2b orbit errors for different software (unit: m).

Software	Orbit		
	Radial	Along	Cross
GKit-SSRDecoder	0.092	0.213	0.381
CSSRlib	1.255	0.268	0.388

B. EVALUATION OF THE ACCURACY OF DECODED HAS PRODUCTS

Similarly, the 11-day three-component orbit errors and clock offset errors of GPS and Galileo satellites between HAS and WUM are presented in Fig. 7. The orbit errors of GPS and

Galileo satellites in HAS service are generally within 0.25 m, except that the differences for satellite G25 occasionally exhibit relatively large fluctuations. The accuracy of orbit radial component even reaches the centimeter level, indicating an outstanding orbit accuracy of HAS GPS corrections. Besides, the clock offset errors of both GPS and Galileo satellites are mostly below 0.15 m, showing no significant fluctuations. However, a few jumps and spikes in clock errors are observed particularly for GPS satellites, while HAS clock corrections for Galileo satellites exhibit significantly better stability than GPS satellites.

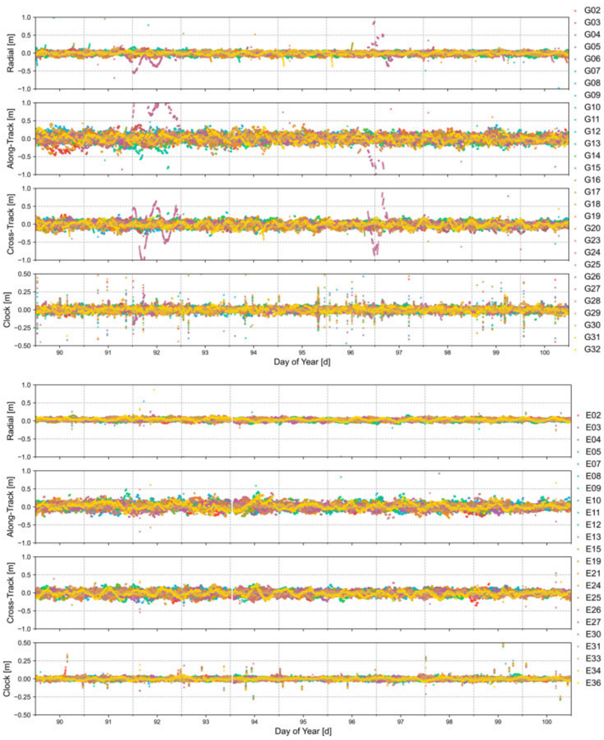

FIGURE 7. HAS orbit and clock errors from DOY 90 to DOY 100 in 2025.

After excluding epochs with significant jumps, the RMS values of HAS orbit errors, clock errors, and SISRE for different satellites over 11 days are concluded in Fig. 8. Within the same constellation, the RMS of orbit and clock offset errors for different satellites are relatively consistent. The GPS orbit and clock offset accuracies are within 0.08 m except for satellite G25, and the accuracy of Galileo satellites is within 0.06 m. The corresponding average values are summarized in Table 6, and the RMS of GPS orbit and clock offset errors are 0.044 m and 0.038 m, while those of Galileo are 0.045 m and 0.023 m, respectively. Overall, the SISRE of GPS and Galileo in HAS service are 0.057 m and 0.048 m, respectively.

TABLE 6. The average RMS of GPS/Galileo HAS orbit, clock, and SISRE errors (unit: m).

Satellite	Orbit	Clock	SISRE
GPS	0.044	0.038	0.057
Galileo	0.045	0.023	0.048

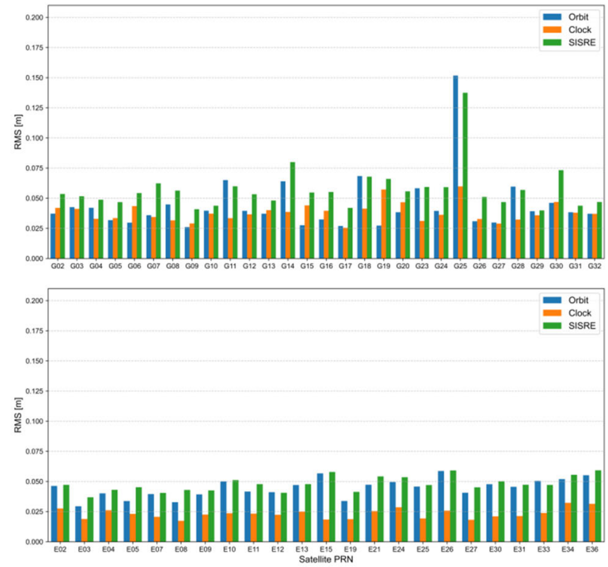

FIGURE 8. RMS values of HAS orbit, clock, and SISRE errors.

TABLE 7. The average difference and standard deviation of HAS code biases (unit: ns).

Signal	GPS			Galileo		
	C2P-C1C	C2L-C1C	C2W-C1C	C5Q-C1C	C7Q-C1C	C6C-C1C
Diff	0.241	0.119	0.224	0.097	0.101	0.094
STD	0.123	0.096	0.152	0.214	0.197	0.146

TABLE 8. Processing strategies for HAS/B2b positioning test.

Items	PPP-B2b service	HAS service
Observation frequency	GPS: L1/L2; BDS: B1I/B3I	GPS:L1/L2; Galileo: E1/E5a
APC correction	IGS20 ANTEX file	IGS14 ANTEX file
Satellite orbits/clocks	Precise ephemeris products from GKit-SSRDecoder	
Satellite DCB	OSB product from GKit-SSRDecoder	
Weighting scheme	Elevation-dependent weight; precision of 3 mm and 0.3 m for raw phase and pseudorange measurements, respectively	
Estimation method	Forward Kalman filter	
Receiver coordinate	Static: estimated as constants kinematic: estimated as white noise process	
Receiver clock	Estimated as white noises per system	
Ionospheric delay	Ionosphere-free combination	
Tropospheric delay	Zenith hydrostatic delays are corrected with the Saastamoinen model, and zenith wet delays are estimated as random walk parameters	
Phase ambiguity	Estimated as float constant	

TABLE 9. The 68th and 90th percentiles of convergence time and positioning errors for PPP-B2b in static solution with all processed stations.

Percentiles	Convergence time (min)		Positioning error (cm)	
	Horizontal direction	Vertical direction	Horizontal direction	Vertical direction
68th	16.5	7.5	6.9	5.7
90th	35.5	20.0	9.1	8.4

The CAS products do not provide OSB correction for the GPS C2P channel, therefore, CODE DCB correction files in which P1P2 and P1C1 DCBs are provided are used

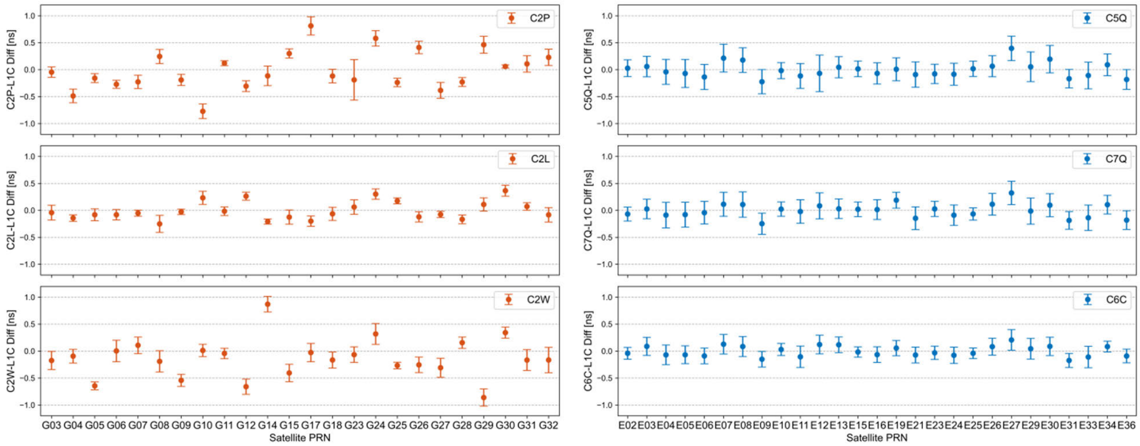

FIGURE 9. Mean difference and standard deviation of HAS code biases relative to post-processed products.

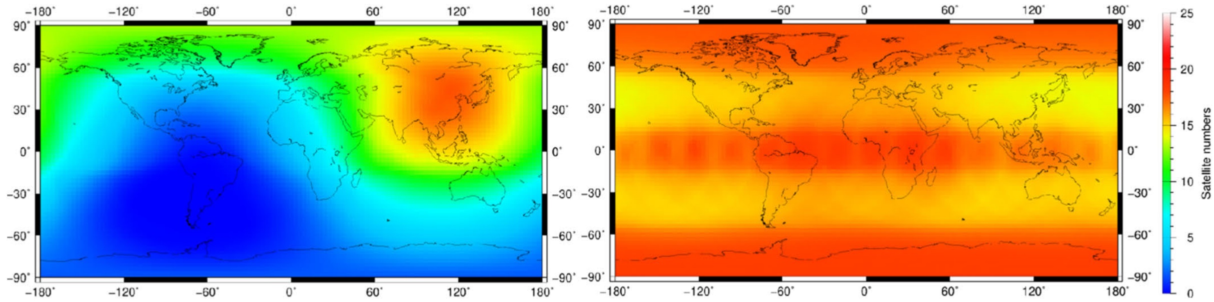

FIGURE 10. Global satellite distribution for PPP-B2b (left)and HAS (right) services.

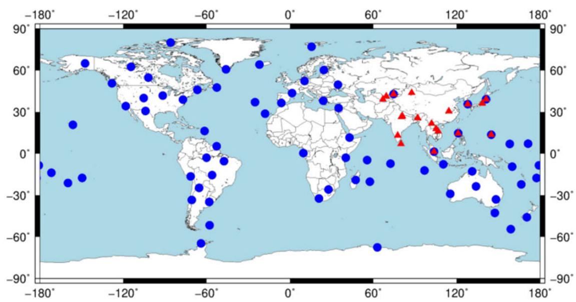

FIGURE 11. Distribution of MGEX stations used for PPP-B2b (red)and HAS (blue) services positioning.

FIGURE 12. Illustration and trajectories of the employed GNSS/IMU equipment setup.

instead, to evaluate the decoded bias products of the GPS C2P channel. Furthermore, the HAS service does not broadcast C2W correction. Given that the current research primarily employs GPS C1C/C2W dual-frequency observations, the

TABLE 10. The 68th and 90th percentiles of convergence time and positioning errors for HAS in static solution with all processed stations.

Percentiles	Convergence time (min)		Positioning error (cm)	
	Horizontal direction	Vertical direction	Horizontal direction	Vertical direction
68^{th}	35.0	12.0	6.4	8.1
90^{th}	62.0	32.0	11.1	13.1

TABLE 11. Real-time PPP Positioning Performance Statistics in kinematic vehicle tests.

Services	Positioning error (cm)			Execution time (s)
	East direction	North direction	Vertical direction	
PPP-B2b	13	11	12	2.649
HAS	18	12	23	29.506

C2P corrections broadcasted by HAS are used as a substitute for C2W signal. The code biases of HAS service are in the OSB format, the HAS DCBs are generated by differencing OSBs. The mean difference and standard deviation of HAS DCB over 11 days are illustrated in Fig. 9 and the corresponding statistical values are shown in Table 7. The differences of C2L-C1C DCBs for GPS satellites are 0.119 ns, outperforming the C2P-C1C and C2W-C1C DCB accuracies which are ∼0.30 ns. The DCB differences of Galileo satellites in HAS service range from 0.094 ns to 0.101 ns, and the C6C-C1C combination has better performance than others. The standard deviation of GPS DCB is lower than that of

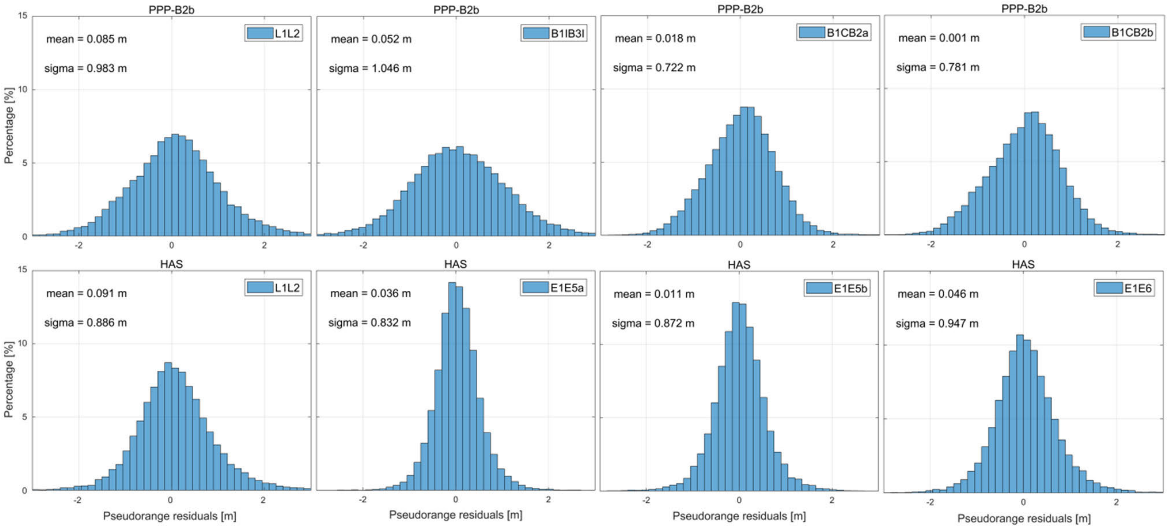

FIGURE 13. Probability distribution of pseudorange residuals for HAS/PPP-B2b services at GAMG station.

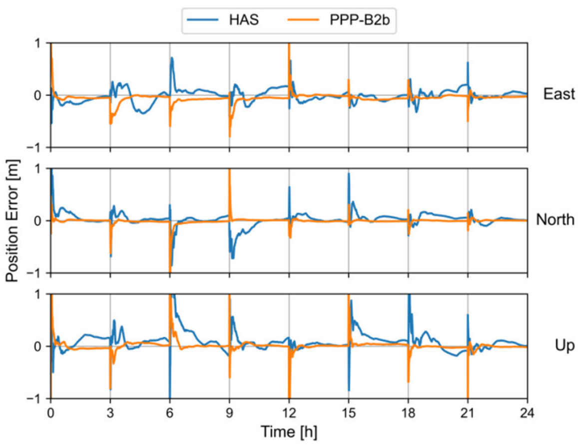

FIGURE 14. Static PPP positioning errors at the GAMG station for PPP-B2b and HAS service.

Galileo, indicating superior stability over several days for GPS DCB.

C. POSITIONING PERFORMANCE VALIDATION USING DECODED B2B AND HAS PRODUCTS

Fig. 10 shows the distribution of satellites providing corrections by PPP-B2b and HAS services over one day, the PPP-B2b service primarily covers China and its surrounding regions (60°E to 160°E, 0°N to 80°N), with up to 20 satellites available in China. The HAS service provides satellite corrections globally, however, the number of available satellites differs across latitudinal regions due to orbital characteristics. Typically, around 20 satellites are available in high- and low-latitude areas while mid-latitude regions have about 15 satellites. The actual usable count may be lower due to factors such as observation environment and satellite elevation angle.

To validate the positioning performance with decoded HAS and B2b products, 70 globally distributed stations (blue dots) together with 20 stations distributed in the Asia-Pacific region (red triangles) from the multi-GNSS experiment (MGEX) were selected to be processed for the static experiment, as shown in Fig. 11. The data duration is from DOY 90 to 100 in 2025 with a sampling rate of 30 s. Concurrently, one set of kinematic data with a sampling rate of 1 s was collected in Xi'an, China, on 9th September, 2025, to assess the performance of kinematic PPP. As illustrated in Fig. 12, the employed equipment setup consists of a multi-band GNSS antenna, a standby antenna, and a MEMS IMU. The vehicle traveled for 5262 s from UTC 01:30:54 to 02:58:42 in an open test area. The lever arms between IMU and the GNSS antenna were precisely calibrated beforehand, and the reference trajectory and attitude were calculated using a post-processing kinematic (PPK)/INS tightly-coupled algorithm using the well-known commercial software Inertial Explorer (IE). All observations were processed with the dual-frequency ionosphere-free combination, and a forward Kalman filter was applied to conduct RT-PPP. The precise ephemeris generated by the GKit-SSRDecoder software is referenced to the satellite CoM, hence, the IGS14 ANTEX file and IGS 20 ANTEX file are applied for HAS and B2b services, respectively. Other detailed strategies are summarized in Table 8. The positioning experiment was conducted using an enhanced version of RTKLIB-b34j software [32], which supports dual-frequency observation with different frequency combinations for multi-GNSS satellites. Additionally, the PPP software incorporates the capability of utilizing OSB corrections. The static PPP experiment was reset at 3-hour intervals to obtain more available solution periods for statistical analysis, i.e., a one-day 24-hour observation file was divided into eight 3-hour sub-files. The horizontal and vertical convergence times are defined as the durations required to achieve and maintain positioning errors below 20 cm and 40 cm, respectively.

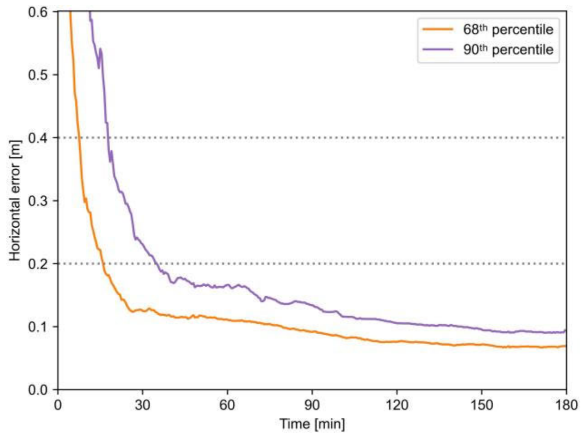

FIGURE 15. Average static PPP horizontal and vertical performance at the 90th and 68th percentiles of PPP-B2b service with all processed stations.

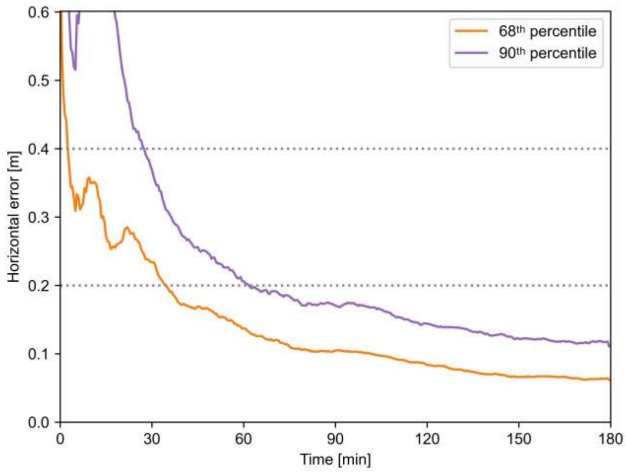

FIGURE 16. Average static PPP horizontal and vertical performance at the 90th and 68th percentiles of HAS service with all processed stations.

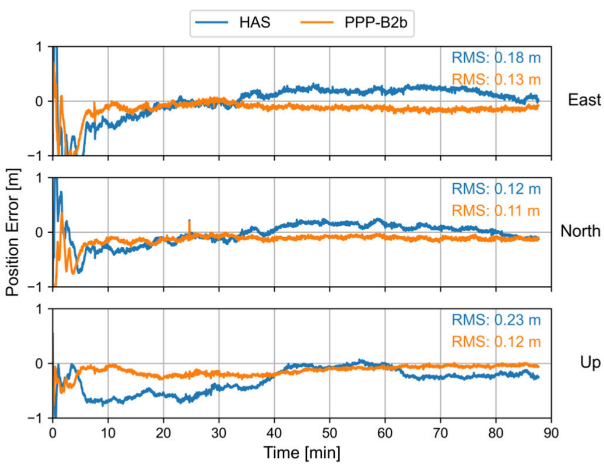

FIGURE 17. Real-time kinematic PPP positioning errors of two services in a vehicle experiment.

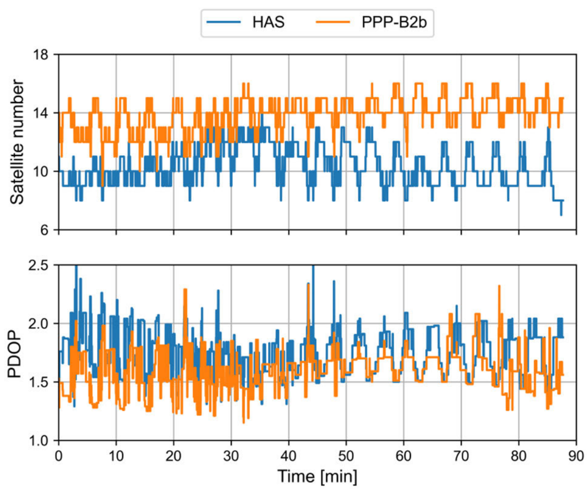

FIGURE 18. Satellite number (top) and PDOP (bottom) for two services in kinematic experiment.

Proper pseudorange bias correction is a critical factor influencing PPP convergence performance [33], Fig. 13 exhibits the probability distribution of pseudorange residuals with different frequency combinations for HAS/PPP-B2b services at GAMG station over one day. With the corrections from the OSB product decoded by the GKit-SSRDecoder, the pseudorange residuals of all frequency combinations follow a zero-mean generalized normal distribution [34], demonstrating the correctness of the code bias decoded from the

GKit-SSRDecoder. For PPP-B2b service, the pseudorange residuals of BDS-3 B1C/B2a and B1C/B2b are within 2 m and are smaller than the GPS L1/L2 and BDS-3 B1I/B3I. With decoded HAS products, the pseudorange residuals of Galileo satellites are significantly smaller than those of GPS satellites, indicating the code bias corrections for Galileo satellites are more accurate than those of GPS satellites. The static PPP positioning errors at the GAMG station are presented in Fig. 14, following each initialization, both services consistently achieve convergence to decimeter and even centimeter-level accuracy. Notably, the PPP-B2b service demonstrates a significantly faster convergence speed and exhibits superior stability in positioning performance.

The time series of average horizontal and vertical static positioning errors at the 90th and 68th percentiles for PPP-B2b service of all processed stations are shown in Fig. 15, and the corresponding convergence time and positioning errors after convergence are provided in Table 9. It takes 35.5 min and 20.0 min to converge in horizontal and vertical directions, respectively, for the 90th percentile positioning errors of PPP-B2b. Meanwhile, the positioning accuracy reaches 9.1 cm and 8.4 cm after convergence. In comparison, the 68th percentile of the convergence time is 16.5 min and 7.5 min in horizontal and vertical directions, with the positioning precision of 6.9 cm and 5.7 cm, respectively.

Similarly, Fig. 16 illustrates the time series of average horizontal and vertical static errors at the 90th and 68th percentiles for HAS service. The corresponding convergence time and positioning errors after convergence are summarized in Table 10. It can be noticed that 90% of HAS PPP solutions need 62.0 min and 32.0 min to converge in horizontal and vertical directions, respectively, while those of the 68th percentile are 35.0 min and 12.0 min. After convergence, the precision of the 68th percentile positioning errors achieved centimeter-level with 6.4 cm and 8.1 cm in horizontal and vertical directions, further demonstrating the corrections decoded by GKit-SSRDecoder are correct and effective.

The kinematic PPP positioning errors for both services are illustrated in Fig. 15, alongside the overall positioning error RMS after 30 min as shown in Table 11. The PPP-B2b service achieves positioning RMS errors of 0.13 m, 0.11 m, and 0.12 m in the East, North, and Up directions, respectively. These results are superior to the corresponding errors of 0.18 m, 0.12 m, and 0.23 m observed for the HAS service. Moreover, the PPP-B2b service not only exhibits faster convergence than HAS but also demonstrates superior positioning stability post-convergence. As further shown in Fig. 16, the HAS service utilizes fewer satellites compared to PPP-B2b, leading to correspondingly higher Position Dilution of Precision (PDOP) values, which likely accounts for its degraded positioning performance. Moreover, Table 11 compares the decoding time for both services over the same 86-minute period. The software decodes the PPP-B2b messages in just 2.649 s, whereas the HAS messages take nearly half a minute. This efficiency disparity is due to the HPVRS decoding approach utilized in the HAS service.

V. CONCLUSION

This study introduces the GKit-SSRDecoder, an open-source C/C++ software designed to bridge the gap between raw PPP-B2b/HAS streams and standardized positioning engines. It provides an end-to-end solution for decoding multi-vendor proprietary formats (e.g., Septentrio, NovAtel, SINO, Unicore) and generating RINEX-compliant precise products. The software supports real-time streaming decoding of orbit, clock, and bias corrections. It features customizable orbit reference centers (i.e., CoM/APC) and standardized output to facilitate seamless integration into existing C/C++ positioning frameworks. The performance was rigorously evaluated using 11 days of Asia-Pacific region station data for PPP-B2b service and global data for HAS service, supplemented by kinematic vehicle tests.

Product analysis shows that the SISRE of BDS-3 and GPS reach 0.086 m and 0.093 m for PPP-B2b service, 0.048 m and 0.057 m for the Galileo and GPS in HAS service, respectively. Besides, the precision of code bias correction is better than 0.52 ns for PPP-B2b service and 0.25 ns for HAS service, with exceptional stability observed in BDS-3 and Galileo products, which exhibit standard deviations of less than 0.14 ns and 0.22 ns, respectively.

Static PPP tests indicate that the PPP-B2b service achieves faster convergence, with the 90th percentile convergence time of 35.5 min and 16.5 min for the horizontal and vertical components, respectively. In comparison, the HAS service requires 62.0 min and 32.0 min to reach the same level. Regarding positioning accuracy after convergence, the 90th percentile for PPP-B2b reaches 9.1 cm horizontally and 8.4 cm vertically, while the corresponding metrics for HAS are 11.1 cm and 13.1 cm. In kinematic vehicle tests, PPP-B2b demonstrates superior robustness due to the increased number of visible satellites, yielding RMS errors of 0.13 m, 0.11 m, and 0.12 m in the East, North, and Up directions, respectively. These results consistently outperform the HAS service, which exhibits corresponding errors of 0.18 m, 0.12 m, and 0.23 m.

Overall, the positioning performance of the Asia-Pacific-oriented PPP-B2b service significantly outperforms the globally-oriented HAS service. This superiority can be attributed to several critical factors: First, the HAS service is currently in its initial operational phase, leading to less stable accuracy in its server-side products, as evidenced by our comparative analysis between HAS and post-processed products. Second, the limited number of HAS monitoring stations globally restricts its ability to provide comprehensive tracking compared to the regional network supporting PPP-B2b. Finally, benefiting from the robust constellation architecture of BDS-3, the satellite visibility and geometry in the Asia-Pacific region are superior to those of the Galileo system. As demonstrated in our kinematic experiments, the increased number of tracked satellites significantly enhances user-side positioning performance and reliability.

Despite its utility, GKit-SSRDecoder has several limitations. First, the software's compatibility with diverse proprietary data formats is currently limited. Second, the current implementation focuses exclusively on PPP-B2b and HAS services and does not yet support other regional or commercial SSR corrections. Moving forward, we plan to expand the range of raw manufacturer data formats supported by the software and incorporate additional high-precision services, such as CLAS and MADOCA. Furthermore, we remain committed to monitoring the evolving landscape of GNSS correction services, ensuring that the software's functional capabilities are promptly synchronized with the latest protocol updates and service enhancements.

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